

Limitations, Ethical Concerns, Future Directions of Machine Learning in Disaster and Pandemic Management

Machine Learning

Briefly, machine learning attempts to make computers and other machines mimic the way in which humans learn. Based on input data which is provided, a computer will predict a pattern. The accuracy of the computer's prediction will be evaluated, and based on this evaluation the algorithm the computer uses will be optimized to reduce error. This process can iterate until a certain accuracy threshold is met (IBM).

General Limitations of Machine Learning

- A major limitation in machine learning is the need for large and unbiased data sets, which can be hard to come by. Additionally it can consume lots of resources.
- Machine learning is prone to error as all algorithms are. Machine learning can produce a logical algorithm based on input data that ends up being inaccurate in making predictions for one reason or another.

Limitations in Disaster and Pandemic Management

- Data must be refined before it is fed to the machine for processing. Considering that massive sets of data are required, the extraction of certain features of the data which is often necessary can prove to be very costly.
 - An example of this that the IEEE brings forth is the use of convolutional neural networks to isolate specific areas into catastrophe zones. Specific forecast features

are required, meaning that they must be isolated from the rest of the data that is provided

- In the use of Earth observational evidence, which gathers information on the Earth's surface, atmosphere, and oceans, intensive parameter tuning is required, and the data needs to be regularly adjusted in order to fit the parameters that are assigned.
- Accuracy can become an issue with machine learning. For one, it is difficult to choose an unbiased parameter through which to gauge accuracy. Furthermore, several parameters have to be set in order to increase accuracy, such as specifying the number of trees in a forest or similar features, which can be very difficult and time consuming.
 - For example, in flood management, there is a complex set of parameters, which includes rainfall, pressure, and rate of flow, and these parameters are interconnected, which can obscure results.
- Applications which are dependent on machine learning, such as sensors for example, must be pre-installed, or they could be susceptible to power outages, especially in the case of a natural disaster. Aside from power outages, they could sustain damage, or transmission lines could be broken.
- Natural disasters and pandemics are inherently unpredictable in nature. Viruses are able to evolve, and natural disasters are becoming more frequent and intense primarily due to climate change. This proves to be a challenge for machine learning in order to keep up with the evolution of these threats.

- In analyzing rainfall and the spread of cholera, Righetto *et. al* developed a model using clustering to predict the spread of disease. However, the weakness in this model was its inability to account for the evolution of new cases.

Ethical Concerns

- The primary ethical concern when collecting data that can be used in machine learning is privacy, as it is heavily dependent on personal information.
 - One area that IEEE discusses for the application of machine learning is in crowd evacuation. The most convenient way to collect data would be to use cameras, but this poses a privacy issue. Hence, different methods must be utilized to analyze crowd density.
 - In general, during a pandemic it is very important to be able to track the health and other important information regarding patients, and to keep a close eye on those affected without infringing on their privacy.
- Related to concerns of privacy are potential security issues, as machine learning systems containing large amounts of data could be susceptible to attacks, which would pose issues to public safety.
 - During the COVID-19 pandemic, there was a breach revealing the sensitive information of over 7000 individuals in Wuhan, China, which led to fraud and created public distress.
 - There is a connection between the occurrence of these events and the prevalence of cybercrime, as cybercriminals look to exploit systems in times of vulnerability,

and there is perhaps no more vulnerable time than when people are quarantined and worrying about their livelihoods.

- In general, public attitudes and trust involving the use of artificial intelligence can vary widely.
 - In spite of its potential in case detection and forecasting the spread of disease as seen during COVID-19, concerns remained over the potential displacement of crucial health care professionals, and if the decisions generated by artificial intelligence could be depended upon.
- While making accurate decisions is the main priority in times of disaster and the use of machine learning could assist with doing so, there is still a fundamental need for human empathy and connection between patients and the healthcare industry, which could be jeopardized with a rapid shift towards machinery. In the long term, a lack of human connection might end up hurting the quality of the care that a patient receives, as the treatment could end up being less comprehensive, and morale would be lowered.

Future Directions

- In the future, models will need to become more adaptable to real-time changes in order to increase their accuracy and provide faster decisions to help more people.
- It is important to enhance public trust in these models, not only through increasing the effectiveness of these models, but also through complete transparency about their input information, what they are intended for, and potential shortcomings.

- The integration of artificial intelligence in the detection and spread of natural disasters and pandemics needs to be balanced with a need for public safety and protection of individual rights

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